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RESEARCH INTERESTS

Self-supervised representation learning and **world models**, especially joint-embedding predictive architectures (JEPA) that learn by predicting in latent space; medical image analysis in low-label regimes; training-free guidance of text-to-image diffusion models.

EDUCATION

University of Dhaka

B.Sc. in Robotics and Mechatronics Engineering • CGPA: 3.20/4.00

Dhaka, Bangladesh

Expected Oct 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Intelligent Systems and Robotics, Digital Image Processing and Robot Vision, Digital Signal Processing, Linear Algebra, Statistics for Engineers.

PUBLICATIONS & MANUSCRIPTS

- **Mind the Gap: A Controlled Evaluation of Slice Separation and Masking in Self-Supervised CT Pretraining.** First author. *Under review* (double-blind).
- **GenMem: Auditing Predicted-Memorability Guidance in Text-to-Image Diffusion.** Undergraduate thesis; *manuscript in preparation*.

RESEARCH EXPERIENCE

Slice Separation and Masking in Self-Supervised CT Pretraining

Jan 2026 – Present

First-author study, University of Dhaka

[🌐 ajmainapon/SSLR](#)

- Asked which design choices matter when adapting **I-JEPA**-style predictive pretraining from natural images to 3D CT; ran a controlled **2×2 study** over context–target slice separation and masking geometry, pretraining a **ViT-B/16** on 1,082 unlabeled TotalSegmentator volumes.
- Found that slice separation, not masking, is the dominant factor, contradicting intuitions carried over from video SSL, and that the two factors interact non-additively; the narrow-gap backbone roughly **doubles linear-probe Dice over random initialization**.
- Evaluated label efficiency (5–100 labeled volumes) against nnU-Net and a matched SwinUNETR comparison, and tested transfer across datasets (AMOS) and modalities (CT→MR, with no MR pretraining).

Senior Thesis — Predicted-Memorability Guidance in Diffusion

Jan 2026 – Present

Advisors: Dr. Md. Mehedi Hasan, Dr. Sejuti Rahman, University of Dhaka

[🌐 ajmainapon/GenMem](#)

- Developed **GenMem**, a training-free method that steers **Stable Diffusion** toward more memorable images by backpropagating a pre-trained memorability predictor through the **DDIM** sampling trajectory, with all model weights frozen.
- Designed the evaluation to separate real effects from proxy gaming: memorability is scored by predictors never used for guidance, and reverse-guidance and random-critic controls test causality; gains are statistically significant with no significant loss in prompt fidelity.
- Characterized where the method breaks: proxy over-optimization in SDXL and three distinct regimes as guidance strength increases.

Undergraduate Research Assistant — Stomatal Phenotyping

2024 – 2025

Funded project, University of Dhaka; paper submitted to Frontiers in Plant Science

[🌐 ajmainapon/StomataAI](#)

- Built a **506-image instance-segmentation benchmark** (6,605 annotated guard-cell complexes and pores) from chickpea leaf microscopy, with locked train/val/test splits.
- Benchmarked YOLO-family and Mask R-CNN segmentation models on identical splits, and automated stomatal density and pore-index extraction, validated against manual counts.
- Related the extracted stomatal traits to drought-stress responses in *C. arietinum* using PCA and hypothesis testing.

HONORS & AWARDS

- **Champion, Insight 2.0 Datathon 2026** (79 teams), IASDS Students' Club, University of Dhaka: lung-cancer survival prediction from SEER registry data. Sep 2026

SKILLS

Programming Python, C++, C

Frameworks PyTorch, timm, MONAI, Hugging Face, torchvision, Ultralytics, Scikit-learn, OpenCV, Nibabel, CUDA

Methods Self-supervised pretraining, diffusion guidance, instance segmentation, controlled ablations, statistical testing

LEADERSHIP & SERVICE

- **President**, Robotics & Mechatronics Student Club, University of Dhaka

Oct 2025 – Present

- **Program Coordinator**, IEEE Robotics & Automation Society, University of Dhaka

Mar 2024 – Apr 2025